

Communication Systems

BRANCH: ELECTRONICS (ECE)

COURSE CODE: EC303

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Communication Systems (EC303) under the Electronics (ECE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Analog Modulation Techniques

Need for modulation. Amplitude Modulation (AM): DSB-FC, DSB-SC, SSB-SC, VSB. Modulators and demodulators. Angle Modulation: Frequency Modulation (FM) and Phase Modulation (PM), Narrowband and Wideband FM.

Unit 2: Pulse Modulation

Sampling theorem, Nyquist rate. Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM), Pulse Position Modulation (PPM). Pulse Code Modulation (PCM), quantization noise, Delta Modulation (DM), ADM.

Unit 3: Digital Modulation & Multiplexing

Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK), QPSK, QAM. Multiplexing: FDM, TDM.

Unit 4: Receiver & Noise Analysis

Superheterodyne receiver, intermediate frequency (IF), image frequency. Thermal noise, shot noise, noise figure, noise temperature. SNR analysis in AM and FM.

AM Modulation Index: $m = (V_{max} - V_{min}) / (V_{max} + V_{min})$. FM Modulation Index: $Beta = \Delta f / f_m$. IF = $f_{LO} - f_s$.

Key Formulas & Exam Pointers

CRITICAL FORMULA / PRINCIPLE:

